

Java Collections and Framework:

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QUESTIONS 8 / 8 completed

Logger

LOGGER

You are working on a project that requires the use of a logging framework. Write a Java program to create an interface called "Logger" with methods for logging different types of messages, such as "debug", "info", "warning", and "error". Implement the interface in a class called "Log4jLogger".

```
1 public class Main {
2
3     public static void main(String[] args) {
4
5         Logger logger = new Log4jLogger();
6
7         logger.debug("Debug message");
8         logger.info("Application started");
9         logger.warning("Low memory warning");
10        logger.error("Database connection failed");
11    }
12 }
13
14 interface Logger {
15
16     void debug(String message);
17     void info(String message);
18     void warning(String message);
19     void error(String message);
20 }
21
```

OUTPUT

DEBUG: Debug message
INFO: Application started
WARNING: Low memory warning
ERROR: Database connection failed

INPUT

Your INPUT goes here!

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QUESTIONS 8 / 8 completed

Character

CHARACTER

You are developing a game that involves different types of characters, such as heroes, villains, and monsters. Write a Java program to create an interface called "Character" with methods for moving, attacking, and taking damage. Implement the interface in classes for each type of character.

```
1 public class Main {
2
3     public static void main(String[] args) {
4
5         Character hero = new Hero();
6         Character villain = new Villain();
7         Character monster = new Monster();
8
9         System.out.println("Hero:");
10        hero.move();
11        hero.attack();
12        hero.takeDamage();
13
14        System.out.println();
15
16        System.out.println("Villain:");
17        villain.move();
18        villain.attack();
19        villain.takeDamage();
20
21        System.out.println();
22    }
23 }

```

OUTPUT

Hero:
Hero moves forward.
Hero attacks with a sword.
Hero takes damage.

INPUT

Your INPUT goes here!

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QUESTIONS 8 / 8 completed

Sort

SORT

Write a Java program to create a list of integers and sort them in ascending order using the Collections framework.

```
1 import java.util.ArrayList;
2 import java.util.Collections;
3 import java.util.List;
4
5 public class SortListExample {
6     public static void main(String[] args) {
7         List<Integer> numbers = new ArrayList<>();
8         numbers.add(45);
9         numbers.add(12);
10        numbers.add(89);
11        numbers.add(5);
12        numbers.add(23);
13        System.out.println("Original List: " + numbers);
14        Collections.sort(numbers);
15        System.out.println("Sorted List (Ascending): " + numbers);
16    }
17 }
18
```

OUTPUT

Original List: [45, 12, 89, 5, 23]
Sorted List (Ascending): [5, 12, 23, 45, 89]

INPUT

Your INPUT goes here!



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QUESTIONS 8 / 8 completed

HashSet

HASHSET
Write a Java program to create a HashSet of strings and remove a specific element from it.

PROGRAM EDITOR

Java Run Save

```
1 import java.util.HashSet;
2
3 public class Main {
4
5     public static void main(String[] args) {
6
7         HashSet<String> names = new HashSet<>();
8
9         names.add("Apple");
10        names.add("Banana");
11        names.add("Mango");
12        names.add("Orange");
13        names.add("Grapes");
14
15        System.out.println("HashSet before removing: " + names);
16
17        names.remove("Mango");
18
19        System.out.println("HashSet after removing Mango: " + names);
20    }
21 }
```

OUTPUT
HashSet before removing: [Apple, Grapes, Mango, Orange, Banana]
HashSet after removing Mango: [Apple, Grapes, Orange, Banana]

INPUT
Your INPUT goes here!



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QUESTIONS 8 / 8 completed

TreeMap

TREEMAP
Write a Java program to create a TreeMap of strings and integers and iterate over its entries.

PROGRAM EDITOR

Java Run Save

```
1 import java.util.Map;
2 import java.util.TreeMap;
3 public class TreeMapSample {
4     public static void main(String[] args) {
5         TreeMap<String, Integer> map = new TreeMap<>();
6         map.put("Banana", 3);
7         map.put("Apple", 5);
8         map.put("Cherry", 8);
9         map.put("Date", 1);
10        System.out.println("TreeMap entries (sorted alphabetically by key):");
11        for (Map.Entry<String, Integer> entry : map.entrySet()) {
12            String key = entry.getKey();
13            Integer value = entry.getValue();
14            System.out.println(key + " => " + value);
15        }
16    }
17 }
18 }
```

OUTPUT
TreeMap entries (sorted alphabetically by key):
Apple => 5
Banana => 3
Cherry => 8

INPUT
Your INPUT goes here!



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QUESTIONS 8 / 8 completed

PriorityQueue

PRIORITYQUEUE
Write a Java program to create a PriorityQueue of integers and add elements to it.

PROGRAM EDITOR

Java Run Save

```
1 import java.util.PriorityQueue;
2
3 public class Main {
4
5     public static void main(String[] args) {
6
7         PriorityQueue<Integer> numbers = new PriorityQueue<>();
8
9         numbers.add(40);
10        numbers.add(10);
11        numbers.add(30);
12        numbers.add(20);
13        numbers.add(50);
14
15        System.out.println("PriorityQueue: " + numbers);
16    }
17 }
```

OUTPUT
PriorityQueue: [10, 20, 30, 40, 50]

INPUT
Your INPUT goes here!



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QUESTIONS 8 / 8 completed

LinkedHashMap

LINKEDHASHMAP
P

Write a Java program to create a LinkedHashMap of strings and integers and retrieve the keys in the order they were inserted.

PROGRAM EDITOR Java Run Save

```
1 import java.util.LinkedHashMap;
2 import java.util.Set;
3
4 public class Main {
5
6     public static void main(String[] args) {
7
8         LinkedHashMap<String, Integer> map = new LinkedHashMap<>();
9
10        map.put("Apple", 10);
11        map.put("Banana", 20);
12        map.put("Mango", 30);
13        map.put("Orange", 40);
14        map.put("Grapes", 50);
15
16        Set<String> keys = map.keySet();
17
18        System.out.println("Keys in insertion order:");
19
20        for (String key : keys) {
21            System.out.println(key);
22        }
23    }
24 }
```

OUTPUT

Keys in insertion order:
Apple
Banana
Mango

INPUT

Your INPUT goes here!



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QUESTIONS 8 / 8 completed

Max-Min

MAX-MIN

Write a Java program to create a Set of integers and perform different operations on it, such as finding the maximum and minimum values.

PROGRAM EDITOR Java Run Save

```
1 import java.util.Set;
2 import java.util.HashSet;
3 import java.util.Collections;
4
5 public class Main {
6
7     public static void main(String[] args) {
8
9         Set<Integer> numbers = new HashSet<>();
10
11        numbers.add(10);
12        numbers.add(25);
13        numbers.add(5);
14        numbers.add(40);
15        numbers.add(15);
16
17        System.out.println("Set: " + numbers);
18
19        int maximum = Collections.max(numbers);
20        int minimum = Collections.min(numbers);
21
22    }
23 }
```

OUTPUT

Set: [5, 40, 25, 10, 15]
Maximum value: 40
Minimum value: 5

INPUT

Your INPUT goes here!